BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, GUJRANWALA

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

PHYSICS

Past Paper 2024 • Paper Code: 6477 / 6478

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics (Intermediate Part-I) Paper : I (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6477

Note: Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. 1 - A 2 Kg mass is placed on the floor of an elevator which is moving downward with 4.9 m/s^2 acceleration, the reaction of floor on the mass is

(A) 9.8 N
(B) 0 N
(C) 4.9 N
(D) 14.7 N

Q2. Which pair of angles gives same range of projectile thrown with velocity v_i

(A) (20°, 60°)
(B) (20°, 40°)
(C) (30°, 60°)
(D) (30°, 70°)

Q3. For circular motion with constant speed v, ω and r are at

(A) 90° with each other
(B) 120° with each other
(C) 60° with each other
(D) 30° with each other

Q4. Which of the following is correct

(A) in irreversible process entropy remains constant
(B) in reversible process entropy increases
(C) in reversible process entropy remains constant
(D) in irreversible process entropy decreases

Q5. To observe one hundred fringes in Michelson's interferometer, the distance travelled by moveable mirror will be minimum in case of ____ light.

(A) Red
(B) Green
(C) Blue
(D) Yellow

Q6. A body in SHM with amplitude x_o goes from mean position to $x_o/2$. Its phase is

(A) 30°
(B) 45°
(C) 60°
(D) 90°

Q7. $\hat{i}\cdot(\hat{j}\times\hat{k})$ = ____

(A) 0
(B) 1
(C) $\hat{i}$
(D) $\hat{j}$

Q8. Two masses 2 Kg and 3 Kg are moving towards each other with velocity 3 m/s and 2 m/s. The total momentum of the system is

(A) 12 Ns
(B) 0 Ns
(C) 13 Ns
(D) −12 Ns

Q9. Mass is a highly concentrated form of ____.

(A) Momentum
(B) Inertia
(C) Energy
(D) Acceleration

Q10. A spectrometer is not used to

(A) study spectrum of light
(B) measure refractive index of material of prism
(C) study polarization of light
(D) measure wavelength of light

Q11. If frequency of stationary waves are increased to higher harmonic which of the following decreases

(A) speed
(B) wavelength
(C) tension in the string
(D) density of string

Q12. Which is renewable source of energy

(A) Biomass
(B) Coal
(C) Oil
(D) Uranium

Q13. Heat is transferred slowly to a gas in a cylinder, the piston is pushed up through 4.0 cm at constant pressure of $8000 Nm^{-2}$. If cross-sectional area of the piston is $0.10 m^2$, work done by the gas is

(A) 32 j
(B) 64 j
(C) 16 j
(D) 96 j

Q14. The complete requirement for a body to be in equilibrium are

(A) $\Sigma \vec{F}=0$ (B) $\Sigma \vec{F}_x=0$

(C) $\Sigma \tau=0$ (D) $\Sigma \vec{F}=0$ and $\Sigma \tau=0$

Q15. If the percentage uncertainty in the radius of a sphere is 3%, then percentage uncertainty in its area is

(A) 3% (B) 6%

(C) 9% (D) 4%

Q16. Two points in a wave $\lambda/4$ distance apart have phase difference

(A) π (B) $\pi/2$

(C) $\pi/3$ (D) 2π

Q17. Bernoulli's equation relates to

(A) pressure, speed and height (B) pressure, force and height

(C) force, speed and pressure (D) force, height and speed

Physics (Intermediate Part-I) Paper : I (Subjective Type)

Time Allowed: 2.40 hours | **Maximum Marks:** 68

> Note: Section I is compulsory. Attempt any THREE (3) questions from Section II.

Q2. (Marks 16) Write short answers to any EIGHT questions. (2 x 8 = 16)

i. What are significant figures? What is rule when first digit dropped is less than 5 while rounding off the data?

ii. What is absolute uncertainty? What is its value?

iii. Why do we find it useful to have two units for the amount of substance, the kilogram and the mole?

iv. Give the drawbacks to use the period of a pendulum as a time standard?

v. Can a vector have a component greater than the vector's magnitude? Explain.

vi. If $\vec{A}+\vec{B}=\vec{O}$, what can you say about the components of the two vectors?

vii. What is position vector? Explain briefly.

viii. Discuss and draw the velocity time graph when car moves with constant acceleration?

ix. Explain the circumstances in which velocity $\bar{v}$ and acceleration $\bar{a}$ are (i) perpendicular to each other (ii) anti-parallel

x. What will happen when a light body collides with a massive body at rest in an elastic collision?

xi. A 70 kg man runs up a long flight of stairs in 4.0 s. The vertical height of the stairs is 10 m. Calculate his power output in watts.

xii. Calculate the work done in kilo joules in lifting a mass of 10 kg (at a steady velocity) through a vertical height of 10 m.

Q3. (Marks 16) Write short answers to any EIGHT questions. (2 x 8 = 16)

i. If a lift is falling freely under gravity, how weightlessness is produced. Use mathematical equations to support your answer.

ii. How do you create a gravity free system?

iii. What is meant by centripetal force and why it must be furnished to an object, if the object is to follow a circular path?

iv. What is meant by moment of inertia with its physical significance? Use equations to support your answer.

v. How Bernoulli's equation is reduced? When a) height difference is negligible b) velocity is constant.

vi. What do you understand by the term viscosity? Also give its unit.

vii. Define damping process. Use a graph to support your answer.

viii. If a mass spring system is hung vertically and set into oscillations, why does the motion eventually stop?

ix. Explain the relation between total energy, potential energy and kinetic energy of a body oscillating with S.H.M

x. How Doppler Effect is used to monitor blood flow? Use diagrammatic explanation to support your answer.

xi. Explain why sound travels faster in warm air than in cold air?

xii. How are beats useful in tuning musical instrument?

Q4. (Marks 12) Write short answers to any SIX questions. (2 x 6 = 12)

i. Under what conditions two or more sources of light behave as coherent sources?

ii. What are the conditions for detectable interference?

iii. 10000 lines per centimeter has been ruled on diffraction grating. Find its grating element.

iv. Why would it be advantageous to use blue light with a compound microscope?

v. Why is meant by " least distance of distinct vision "'?

vi. Find magnifying power of convex lens of 25 cm focal length acts as a magnifying glass.

vii. Does entropy of a system increase or decrease due to friction?

viii. Is it possible to construct a heat engine that will not expel heat into the atmosphere?

ix. Derive Charles' law from Kinetic theory of gases

> Note: SECTION – II

Q5.

(a) Does the inertia depend on the momentum of a body? Give its reason. Also state and explain the law of conservation of linear momentum. [5]

(b) Show that the three vectors $\hat{i}+\hat{j}+\hat{k}$, $2\hat{i}-3\hat{j}+\hat{k}$ and $4\hat{i}+\hat{j}-5\hat{k}$ are mutually perpendicular. [3]

Q6.

(a) Stationary waves are also called standing waves, why? Discuss stationary waves in air column of an open organ pipe. [5]

(b) How large a force is required to accelerate an electron (m = 9.1 x 10^{-31}kg) from rest to a speed of 2 x $10^{7}$$ms^{-1}$ through a distance of 5cm? [3]

Q7.

(a) How does a space satellite acquire an artificial gravity? [5]

(b) A block weighing 4.0 Kg extends a spring by 0.16 m from its unstretched position. If the block is removed and 0.50 kg body is hung from same spring, now what is its period of vibration? [3]

Q8.

(a) Explain four stroke petrol engine in detail. What is the efficiency of a diesel engine? [5]

(b) Water flows through a hose, whose internal diameter is 1cm at a speed of 1 ms^{-1}. What should be the diameter of the nozzle if the water is to emerge at 21ms^{-1}? [3]

Q9.

(a) What is meant by diffraction of light? Also discuss the diffraction of light through a narrow slit. [5]

(b) An astronomical telescope having magnifying power of 5 consists of two thin lenses 24 cm apart. Find the focal lengths of the lenses. [3]

GROUP – II

Physics (Intermediate Part-I) Paper : I (Objective Type)

Time Allowed: 20 Minutes	**Maximum Marks:** 17	**Paper Code:** 6478

> Note: Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. 1 - In angular motion, the centripetal force is

(A) $mr^2\omega^2$ (B) $m^2r^2\omega$

(C) $mr^2\omega$ (D) $mr\omega^2$

Q2. If temperature of sink increases, the efficiency of Carnot Engine

(A) decreases (B) increases

(C) remains the same (D) first increases then decreases

Q3. The detector in photo-phone is made up of

(A) Germanium (B) Selenium

(C) Cadmium (D) Silicon

Q4. The dimensions of the relation $\sqrt{((F \times l)/m)}$ are equal to the dimensions of

(A) Force (B) Impulse

(C) Momentum (D) Velocity

Q5. Dot product of force and velocity is

(A) Work (B) Momentum

(C) Power (D) Impulse

Q6. In reversible process the entropy of system

(A) increases (B) decreases

(C) remains constant (D) becomes zero

Q7. Newton rings are formed due to

(A) diffraction (B) reflection

(C) refraction (D) interference

Q8. The maximum drag force on falling sphere is 9.8 N, its weight is

(A) 9.8 N (B) 19.8 N

(C) 4.9 N (D) 49 N

Q9. Distance covered by a body in one vibration is 20 cm. The amplitude of vibration will be

(A) 5 cm (B) 10 cm

(C) 15 cm (D) 20 cm

Q10. Torque is the rotational analogous of

(A) Momentum (B) Impulse

(C) Force (D) Power

Q11. In which quadrant, vector $3\hat{i}-5\hat{j}$ lies?

(A) 1st (B) 2nd

(C) 3rd (D) 4th

Q12. A fighter plane is chasing another plane, when it opens fire, its speed

(A) increases (B) decreases

(C) remains constant (D) first increases then decreases

Q13. 2 revolutions are equal to

(A) π/2 rad (B) π rad

(C) 2π rad (D) 4π rad

Q14. Speed of sound is independent of

(A) density (B) temperature

(C) elasticity (D) pressure

Q15. The unit of work in base units is

(A) kg ms^2 (B) kg m^2s^{-2}

(C) kg $m^{-2}s^2$ (D) kg m^2s^2

Q16. Star moving towards the earth shows

(A) red shift (B) blue shift

(C) yellow shift (D) green shift

Q17. The distance covered by free falling body in 2 seconds is

(A) 9.8 m (B) 19.6 m

(C) 4.9 m (D) 49 m

Physics (Intermediate Part-I) Paper : I (Subjective Type)

Time Allowed: 2.40 hours | **Maximum Marks:** 68

Note: Section I is compulsory. Attempt any THREE (3) questions from Section II.

Q2. (Marks 16) Write short answers to any EIGHT questions. (2 x 8 = 16)

i. The period of pendulum is measured by a stop watch. What types of errors are possible in the time standard?

ii. Does a dimensional analysis give any information on constant of proportionality that may appear in algebraic expressions? Explain.

iii. Differentiate between precision and accuracy.

iv. How many seconds are there in one year? Explain.

v. Can a vector have a component greater than the vector's magnitude?

vi. A force of 10 N makes an angle of 60° with x-axis. Find its x and y components.

vii. Give two factors on which turning effect depends.

viii. Explain the circumstances in which velocity $\vec{v}$ and acceleration $\vec{a}$ are perpendicular to one another.

ix. A rubber ball and a lead ball of same size are moving with same velocity. Which ball has greater momentum and why?

x. How will you differentiate between uniform and variable velocity?

xi. An object has 1 J of potential energy. Explain.

xii. What is escape velocity? Write the formula of escape velocity.

Q3. (Marks 16) Write short answers to any EIGHT questions. (2 x 8 = 16)

i. A disc and a hoop start moving down from the top of an inclined plane at the same time. Which one will be moving faster on reaching the bottom?

ii. Why centripetal force is required to keep a body moving on a circular track?

iii. State the direction of the following vectors in simple situations: angular momentum and angular velocity

iv. What does (INTELSAT) stand for?

v. Explain the term viscosity.

vi. What is difference between laminar flow and turbulent flow?

vii. Does frequency depend on amplitude for harmonic oscillator?

viii. Differentiate between undamped and damped oscillations with the help of a graph between amplitude and time.

ix. Name two characteristics of simple harmonic oscillator.

x. As a result of a distant explosion, an observer senses a ground tremor and then hears the explosion. Explain the time difference.

xi. How are beats useful in tuning musical instruments?

xii. How bats navigate their food?

Q4. (Marks 12) Write short answers to any SIX questions. (2 x 6 = 12)

i. An oil film spreading over a wet footpath shows colours. Explain.

ii. How will you differentiate between interference and diffraction of light waves?

iii. 20000 lines per centimeter has been ruled on a diffraction grating. Find its grating element.

iv. How the power is lost in optical fibre through dispersion? Explain.

v. Why would it be advantageous to use blue light with a compound microscope?

vi. Find magnifying power of convex lens of 15 cm focal length acts as a magnifying glass.

vii. Specific heat of a gas at constant pressure is greater than specific heat at constant volume. Why?

viii. Why is the average velocity of the molecules in a gas zero but the average of the square of velocities is not zero?

ix. State Second Law of Thermodynamics in terms of entropy.

> Note: SECTION – II

Q5.

(a) What is an isolated system? State and explain law of conservation of linear momentum. [5]

(b) Given that $\vec{A}=\hat{i}-2\hat{j}+3\hat{k}$ and $\vec{B}=3\hat{i}-4\hat{k}$, find the projection of $\vec{A}$ and $\vec{B}$ [3]

Q6.

(a) Discuss the interconversion of potential and kinetic energy when frictional force is not considered. [5]

(b) The wavelength of the signals from a radio transmitter is 1500 m and the frequency is 200 KHz. What is wavelength for a transmitter operating at 1000 KHz and with what speed the radio waves travel? [3]

Q7.

(a) What is meant by real and apparent weight? Develop a relation between real and apparent weight (in case of an elevator). [5]

(b) What should be length of a simple pendulum whose period is 1.0 second at a place where g = 9.8 ms^{-2} [3]

Q8.

(a) Derive Bernoulli's Equation for an ideal fluid. [5]

(b) 336 J of energy is required to melt 1 g of ice at 0°C. What is the change in entropy of 30 g of water at 0°C as it is changed to ice at 0°C by a refrigerator? [3]

Q9.

(a) What is Michelson's interferometer? Explain its working and derive its equation. [5]

(b) A glass light pipe in air will totally internally reflect a light ray if its angle of incidence is at least 39°. What is the minimum angle for total internal reflection if pipe is in water (Refractive index of water = 1.33) [3]